Supplementary Information for: Probabilistic Design and Safety-Constrained Optimization of a Firewater-Powered Portable Valve Actuation System

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S1. Reproduction

Software: HydraTorque_MC.py version 2.1.0 (numpy and matplotlib only), archived at https://doi.org/10.5281/zenodo.23969346. Commands: python HydraTorque_MC.py --demo (verification suite, about one minute); python -m pytest -q tests (automated checks); python HydraTorque_MC.py --reproduce --outdir results (every result of the paper, seed 2026, about 12 minutes); python make_figures.py (figures); python manuscript/src/build_manuscript.py (manuscript and this document); python HydraTorque_MC.py --legacy (the concept-stage study, unchanged, in legacy_v1/).

S2. Model switches and defaults

Switch	Model 2 (on)	Concept-stage model (off)
flow_supply	Source pressure falls with flow, p_s - (p_s - 20 psi)(Q/Q_a)^1.85	Constant supply pressure
foul_flow	Strainer and fouling loss proportional to Q^2 (sampled value at 60 gpm)	Constant loss in psi
back	Exhaust-hose back-pressure, 2.0 (Q/100)^2 (L_d/100) psi	None
start_eff	Breakaway and stall efficiency lower than running efficiency by 0.05 to 0.20	Same efficiency
leak	Motor leakage proportional to the pressure drop	Constant volumetric factor
surge	Joukowsky surge on stall with relief override and unrelieved fraction	Stall at the relief pressure, no surge

Table S1. The six mechanisms of model 2 and the concept-stage behavior they replace.

S3. Water, flow limit and reserve

Limit (gpm)	Success	Safe stroke	In time	Interference v1	Reserve violated	Safe, adaptive	Water (L)
15	17.7%	16.0%	21.4%	0.1%	0.00%	16.0%	734
20	33.4%	29.6%	40.0%	0.4%	0.00%	29.6%	835
30	40.1%	34.7%	49.2%	3.3%	0.00%	34.7%	888
40	58.3%	50.1%	73.1%	10.1%	0.01%	50.1%	1,032
60	75.5%	62.7%	95.8%	31.6%	0.18%	62.7%	1,609
80	76.2%	60.4%	96.6%	51.8%	0.68%	60.4%	1,622
100	76.2%	58.0%	96.6%	64.8%	1.31%	58.0%	1,622
150	76.2%	54.7%	96.6%	75.0%	2.66%	54.8%	1,621

Table S2. Flow-limit sweep for the baseline tool (60,000 draws): success, safe stroke, stroke within 15 minutes, interference (v1 metric), reserve violation, safe stroke with the adaptive rule and median water per successful stroke.

Spare-flow multiple	Reserve violated	Safe stroke, no rule	Safe stroke, adaptive	In time, adaptive
0.10	55.4%	62.3%	48.2%	71.1%
0.25	21.1%	62.3%	60.0%	91.2%
0.50	3.6%	62.3%	62.1%	95.5%
1.00	0.2%	62.3%	62.3%	95.9%
2.00	0.0%	62.3%	62.3%	95.9%

Table S3. Spare flow of shared sources as a multiple of the prior (60 gpm baseline).

S4. Surge, accumulator and chart

Tool	Surge multiplier	Over-torque	Damage	Safe stroke
baseline	0.0	4.9%	0.08%	71.5%
baseline	0.5	9.5%	0.33%	67.6%
baseline	1.0	15.2%	1.12%	62.9%
baseline	1.5	20.7%	2.48%	58.4%
baseline	2.0	25.7%	4.15%	54.3%
baseline	3.0	34.4%	8.10%	47.3%
baseline + accumulator 0.75	0.0	4.9%	0.08%	71.5%
baseline + accumulator 0.75	0.5	5.8%	0.12%	70.7%
baseline + accumulator 0.75	1.0	6.9%	0.18%	69.7%
baseline + accumulator 0.75	1.5	8.2%	0.24%	68.7%
baseline + accumulator 0.75	2.0	9.5%	0.33%	67.6%
baseline + accumulator 0.75	3.0	12.3%	0.62%	65.2%
optimized allowable|0.75	0.0	0.5%	0.00%	60.8%
optimized allowable|0.75	0.5	0.8%	0.01%	60.6%
optimized allowable|0.75	1.0	1.0%	0.01%	60.4%
optimized allowable|0.75	1.5	1.4%	0.02%	60.1%
optimized allowable|0.75	2.0	1.9%	0.03%	59.7%
optimized allowable|0.75	3.0	3.2%	0.11%	58.8%

Table S4. Over-torque, damage and safe stroke against the surge multiplier (0 = no surge).

Basis	Multiple	Success	Safe stroke	Over-torque	Damage
median	0.80	19.9%	19.7%	0.9%	0.04%
median	1.00	33.5%	32.9%	1.4%	0.07%
median	1.25	49.0%	47.5%	2.7%	0.16%
median	1.50	61.1%	57.5%	5.1%	0.32%
median	2.00	75.5%	62.7%	15.2%	1.11%
median	2.50	81.8%	52.9%	32.2%	3.41%
median	3.00	84.5%	39.6%	48.5%	7.61%
median	4.00	86.0%	23.7%	65.7%	20.42%
allowable	0.20	8.3%	8.2%	0.6%	0.02%
allowable	0.25	16.7%	16.6%	0.8%	0.04%
allowable	0.30	26.7%	26.3%	1.2%	0.06%
allowable	0.40	46.1%	44.8%	2.4%	0.13%
allowable	0.50	61.1%	57.5%	5.1%	0.32%
allowable	0.60	71.0%	62.9%	10.1%	0.72%
allowable	0.70	77.2%	61.7%	18.2%	1.47%
allowable	0.80	81.0%	55.5%	28.6%	2.83%
allowable	1.00	84.5%	39.6%	48.5%	7.61%

Table S5. Relief-setting sweep, baseline tool (chart multiple of the class median breakaway torque, or of the nominal allowable stem torque).

S5. Optimization details

Accum.	Start	End	Turns	Success	Safe	Over-torque	Damage	In time	Time	Water
0.00	0	0	1.0 + 2.0	75.5%	62.7%	15.2%	1.11%	95.8%	7.1	1,609
0.00	0	40	1.0 + 2.0	75.2%	65.6%	12.0%	0.61%	95.4%	7.3	1,603
0.00	0	20	1.0 + 2.0	74.1%	67.6%	8.9%	0.34%	94.0%	7.9	1,583
0.00	0	10	1.0 + 2.0	71.4%	66.4%	7.6%	0.28%	90.0%	9.0	1,528
0.00	20	0	1.0 + 2.0	74.8%	62.1%	14.0%	0.98%	94.9%	7.5	1,597
0.00	10	0	1.0 + 2.0	73.8%	61.2%	13.8%	0.97%	93.4%	8.1	1,576
0.00	20	20	1.0 + 2.0	73.4%	67.0%	7.8%	0.21%	93.0%	8.2	1,570
0.00	10	10	1.0 + 2.0	68.0%	63.1%	6.2%	0.15%	85.2%	9.3	1,343
0.00	10	10	0.5 + 1.0	72.9%	67.8%	6.2%	0.15%	92.3%	8.5	1,560
0.00	10	10	2.0 + 4.0	45.9%	42.3%	6.2%	0.15%	56.2%	10.7	935

Accum.	Start	End	Turns	Success	Safe	Over-torque	Damage	In time	Time	Water
0.00	10	10	4.0 + 8.0	31.3%	28.6%	6.2%	0.15%	37.4%	12.5	819
0.75	0	0	1.0 + 2.0	75.5%	69.6%	6.9%	0.18%	95.8%	7.1	1,609
0.75	0	40	1.0 + 2.0	75.2%	69.9%	6.3%	0.15%	95.4%	7.3	1,603
0.75	0	20	1.0 + 2.0	74.1%	69.5%	5.7%	0.12%	94.0%	7.9	1,583
0.75	0	10	1.0 + 2.0	71.4%	67.2%	5.4%	0.11%	90.0%	9.0	1,528
0.75	20	0	1.0 + 2.0	74.8%	69.0%	6.7%	0.18%	94.9%	7.5	1,597
0.75	10	0	1.0 + 2.0	73.8%	68.0%	6.7%	0.18%	93.4%	8.1	1,576
0.75	20	20	1.0 + 2.0	73.4%	68.9%	5.5%	0.12%	93.0%	8.2	1,570
0.75	10	10	1.0 + 2.0	68.0%	63.9%	5.2%	0.11%	85.2%	9.3	1,343
0.75	10	10	0.5 + 1.0	72.9%	68.7%	5.2%	0.11%	92.3%	8.5	1,560
0.75	10	10	2.0 + 4.0	45.9%	43.0%	5.2%	0.11%	56.2%	10.7	935
0.75	10	10	4.0 + 8.0	31.3%	29.2%	5.2%	0.11%	37.4%	12.5	819

Table S5b. Creep study, all cases (baseline tool): accumulator fraction, creep flows (gpm) at the start and at the end of travel, turns at those flows, success, safe stroke, over-torque, damage, strokes in time, median time (min) and water (L).

Family	Feasible	Non-dominated	D	R	Relief	Q limit	Creep	Safe	Over-torque	Water (L)
allowable|0	418	231	305	100	0.50	33	none	54.0%	0.99%	926
median|0	316	204	203	148	1.51	29	none	51.4%	0.94%	659
allowable|0.75	690	415	81	438	0.52	76	none	60.0%	0.93%	1,136
median|0.75	537	355	141	217	1.58	75	none	58.5%	0.90%	960
allowable|0|creep	541	312	270	160	0.51	70	30 / 5	59.5%	0.88%	1,379
allowable|0.75|creep	732	431	77	554	0.54	96	26 / 22	63.9%	0.98%	1,389
median|0|creep	418	277	392	100	1.56	59	16 / 7	60.5%	0.80%	1,251
median|0.75|creep	554	391	447	115	1.58	73	33 / 16	63.8%	0.87%	1,689

Table S6. Search results by family (basis|accumulator fraction): feasible designs under the 1% constraint, size of the non-dominated set, and the best feasible design (in-sample).

Class	D	R	Relief	Q limit	Safe	Over-torque
gate_12in	312	159	0.52	19	68.6%	0.83%
gate_24in	144	335	0.52	74	63.1%	0.81%
butterfly_36in	460	82	0.54	88	62.7%	1.00%
plug_16in	289	128	0.54	27	66.9%	1.00%
legacy_seized	316	120	0.93	42	48.7%	0.88%

Table S7. Best design for each valve class separately (allowable-torque chart, accumulator 0.75, P(over-torque) <= 1%).

Family	Worst-case limit	D	R	Relief	Q limit	Worst-case safe	Worst-case over-torque
allowable|0	2%	133	124	0.39	23	12.5%	1.2%
allowable|0	3%	67	244	0.42	28	14.7%	2.1%
allowable|0	5%	229	102	0.43	25	19.0%	3.3%
allowable|0	10%	195	122	0.49	27	23.5%	8.4%
allowable|0.75	2%	157	138	0.42	33	18.2%	2.0%
allowable|0.75	3%	157	138	0.42	33	18.2%	2.0%
allowable|0.75	5%	409	89	0.43	60	23.3%	4.6%
allowable|0.75	10%	135	336	0.48	61	29.3%	9.5%
allowable|0|creep	2%	193	127	0.41	22	15.2%	1.9%
allowable|0|creep	3%	191	138	0.43	38	21.0%	3.0%
allowable|0|creep	5%	274	139	0.43	92	23.1%	4.2%
allowable|0|creep	10%	90	367	0.49	97	28.1%	8.6%
allowable|0.75|creep	2%	117	207	0.41	74	18.5%	1.8%
allowable|0.75|creep	3%	374	70	0.42	80	19.4%	2.3%
allowable|0.75|creep	5%	305	96	0.44	71	23.3%	3.4%
allowable|0.75|creep	10%	360	104	0.51	88	30.7%	10.0%

Table S8. Robust selection: for each limit on the worst-case over-torque probability over the assumption shifts, the candidate with the largest worst-case safe-stroke probability.

S6. Valve classes with unequal stem strength

Family	Safe stroke (mix)	Gate 12 in	Gate 24 in	Butterfly 36 in	Plug 16 in	Legacy seized
allowable|0.75	60.3%	1.4%	0.8%	0.8%	0.9%	0.7%
median|0.75	56.0%	0.1%	0.6%	3.5%	0.2%	0.9%
allowable|0.75|creep	62.3%	1.0%	0.8%	0.9%	0.9%	0.8%
median|0.75|creep	59.2%	0.8%	0.8%	1.8%	1.6%	0.3%

Table S9. Best design of each chart basis (accumulator 0.75, constraint on the mix) when the allowable stem torques are 4.0, 3.0, 2.2, 3.5 and 2.0 times the class median breakaway torque; over-torque probability by class.

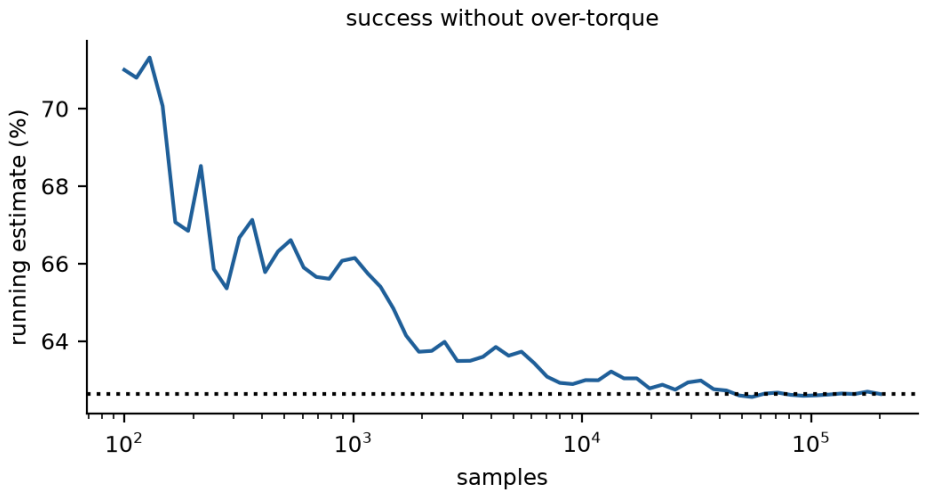


Fig. S1. Convergence of the running estimate of the probability of a safe stroke with the number of samples (baseline tool).

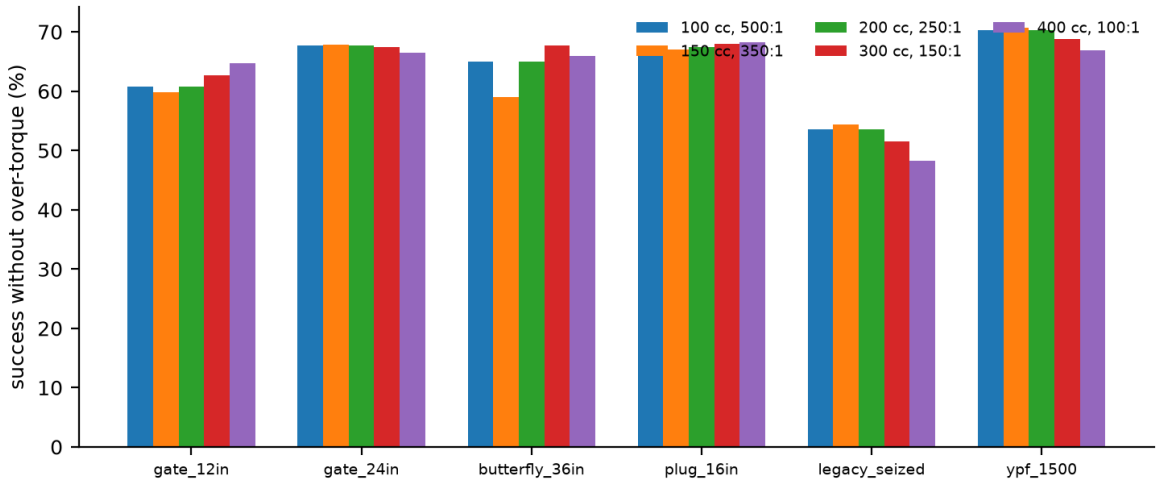


Fig. S2. Probability of a safe stroke by valve class for five displacement and ratio pairs on the ridge, baseline relief chart and flow limit.

S7. Challenge valve and robustness of all designs

Design	Success	Safe stroke	Over-torque	Damage
baseline (chart 2x median, no accumulator)	78.5%	70.2%	9.55%	0.280%
optimized allowable|0	55.0%	54.7%	0.49%	0.005%
optimized median|0	54.9%	54.6%	0.46%	0.003%

Design	Success	Safe stroke	Over-torque	Damage
optimized allowable|0.75	60.7%	60.1%	0.64%	0.005%
optimized median|0.75	57.8%	57.4%	0.58%	0.005%
optimized allowable|0|creep	63.0%	62.7%	0.55%	0.005%
optimized allowable|0.75|creep	66.3%	65.7%	0.83%	0.015%
optimized median|0|creep	62.6%	62.2%	0.58%	0.003%
optimized median|0.75|creep	67.9%	67.3%	0.75%	0.007%
robust allowable|0 (worst-case limit 0.03)	20.1%	20.1%	0.01%	0.000%
robust allowable|0 (worst-case limit 0.1)	45.0%	44.9%	0.16%	0.002%
robust allowable|0.75 (worst-case limit 0.03)	35.8%	35.8%	0.01%	0.000%
robust allowable|0.75 (worst-case limit 0.1)	59.4%	59.2%	0.30%	0.002%
robust allowable|0|creep (worst-case limit 0.03)	43.1%	43.0%	0.05%	0.000%
robust allowable|0|creep (worst-case limit 0.1)	55.8%	55.6%	0.31%	0.003%
robust allowable|0.75|creep (worst-case limit 0.03)	40.6%	40.6%	0.03%	0.000%
robust allowable|0.75|creep (worst-case limit 0.1)	60.5%	60.2%	0.43%	0.003%
baseline + accumulator 0.75	78.5%	73.5%	5.60%	0.118%

Table S10. The 1,500 N.m challenge valve class (lognormal, sigma 0.35, 10% seizure), all designs evaluated.

Design	Safe (out of sample)	Over-torque (o.o.s.)	Worst-case safe	Worst-case over-torque
baseline (chart 2x median, no accumulator)	62.6%	15.20%	33%	53.0%
optimized allowable|0	53.8%	0.99%	25%	12.2%
optimized median|0	51.4%	0.88%	25%	11.6%
optimized allowable|0.75	59.9%	0.99%	30%	13.0%
optimized median|0.75	58.4%	0.93%	28%	12.2%
optimized allowable|0|creep	59.5%	0.87%	30%	12.1%
optimized allowable|0.75|creep	63.7%	1.00%	33%	13.7%
optimized median|0|creep	60.6%	0.79%	31%	11.8%
optimized median|0.75|creep	64.0%	0.91%	33%	13.3%
robust allowable|0 (worst-case limit 0.03)	35.5%	0.06%	14%	2.1%
robust allowable|0 (worst-case limit 0.1)	49.7%	0.51%	22%	8.2%
robust allowable|0.75 (worst-case limit 0.03)	42.1%	0.05%	18%	2.0%
robust allowable|0.75 (worst-case limit 0.1)	57.6%	0.52%	28%	9.3%
robust allowable|0|creep (worst-case limit 0.03)	46.4%	0.09%	20%	3.0%
robust allowable|0|creep (worst-case limit 0.1)	56.4%	0.52%	27%	8.7%
robust allowable|0.75|creep (worst-case limit 0.03)	44.4%	0.05%	19%	2.2%
robust allowable|0.75|creep (worst-case limit 0.1)	59.7%	0.57%	30%	9.8%
baseline + accumulator 0.75	69.5%	6.88%	40%	38.7%

Table S11. All designs: out-of-sample safe stroke and over-torque, and the worst case over the assumption shifts.

S8. Transient check, stem strength and confidence limits

Case	Wave speed	Stopping time	Classical	Simulation
instantaneous	150	0.000	26.00	26.00
instantaneous	250	0.000	43.34	43.34
instantaneous	400	0.000	69.35	69.35
slow (Michaud)	300	0.800	26.00	26.00
slow (Michaud)	300	1.200	17.34	17.34
slow (Michaud)	300	2.000	10.40	10.40
slow (Michaud)	300	4.000	5.20	5.20
short (reduction factor)	300	0.050	52.01	52.01
short (reduction factor)	300	0.200	52.01	52.01

Table S12. The transient simulation against the classical surge relations (2.5-inch hose, 60 gpm, 60 m, reservoir upstream, no relief valve): wave speed (m/s), stopping time (s), classical surge (psi) and simulated surge (psi).

Response time (s)	Median	95th percentile	Above 0.3
0.01	−0.20	0.01	0%
0.03	−0.22	0.00	0%
0.10	−0.28	0.03	1%
0.30	−0.08	0.45	11%
1.00	0.29	0.81	48%

Table S13. Unrelieved fraction of the surge implied by the transient simulation against the response time of the relief valve (120 random cases each): median, 95th percentile and share above the prior bound of 0.3.

Tool	Stem strength	Over-torque	Safe stroke
baseline	0.5	86.31%	7.7%
baseline	0.6	71.22%	18.5%
baseline	0.7	53.26%	32.0%
baseline	0.8	36.90%	44.7%
baseline	0.9	24.16%	55.0%
baseline	1.0	15.30%	62.2%
baseline	1.1	9.34%	67.1%
baseline	1.2	5.75%	70.0%
baseline	1.5	1.29%	73.8%
baseline + accumulator 0.75	0.5	77.50%	13.4%
baseline + accumulator 0.75	0.6	58.43%	27.7%
baseline + accumulator 0.75	0.7	39.13%	42.9%
baseline + accumulator 0.75	0.8	23.52%	55.4%
baseline + accumulator 0.75	0.9	13.03%	64.0%
baseline + accumulator 0.75	1.0	6.78%	69.2%
baseline + accumulator 0.75	1.1	3.38%	72.0%
baseline + accumulator 0.75	1.2	1.64%	73.5%
baseline + accumulator 0.75	1.5	0.15%	74.8%
optimized allowable|0.75	0.5	49.32%	26.4%
optimized allowable|0.75	0.6	27.17%	41.0%
optimized allowable|0.75	0.7	13.19%	50.7%
optimized allowable|0.75	0.8	5.74%	56.1%
optimized allowable|0.75	0.9	2.32%	58.5%
optimized allowable|0.75	1.0	0.88%	59.5%
optimized allowable|0.75	1.1	0.32%	60.0%
optimized allowable|0.75	1.2	0.11%	60.1%
optimized allowable|0.75	1.5	0.00%	60.2%
optimized allowable|0.75|creep	0.5	51.74%	27.9%
optimized allowable|0.75|creep	0.6	28.87%	43.5%
optimized allowable|0.75|creep	0.7	13.89%	54.0%
optimized allowable|0.75|creep	0.8	5.99%	59.8%
optimized allowable|0.75|creep	0.9	2.38%	62.4%
optimized allowable|0.75|creep	1.0	0.89%	63.5%
optimized allowable|0.75|creep	1.1	0.32%	63.9%
optimized allowable|0.75|creep	1.2	0.11%	64.1%
optimized allowable|0.75|creep	1.5	0.01%	64.1%

Table S14. Over-torque and safe-stroke probability against the true stem strength relative to the data-sheet value (60,000 draws).

Design	Over-torque	Lower	Upper
baseline (chart 2x median, no accumulator)	15.204%	15.047%	15.362%
optimized allowable|0	0.988%	0.945%	1.032%
optimized median|0	0.878%	0.839%	0.920%
optimized allowable|0.75	0.993%	0.951%	1.038%
optimized median|0.75	0.932%	0.891%	0.976%
optimized allowable|0|creep	0.867%	0.827%	0.909%
optimized allowable|0.75|creep	1.004%	0.961%	1.049%
optimized median|0|creep	0.787%	0.749%	0.826%

Design	Over-torque	Lower	Upper
optimized median|0.75|creep	0.907%	0.867%	0.950%
robust allowable|0 (worst-case limit 0.03)	0.064%	0.053%	0.076%
robust allowable|0 (worst-case limit 0.1)	0.509%	0.479%	0.541%
robust allowable|0.75 (worst-case limit 0.03)	0.045%	0.037%	0.056%
robust allowable|0.75 (worst-case limit 0.1)	0.515%	0.485%	0.548%
robust allowable|0|creep (worst-case limit 0.03)	0.090%	0.078%	0.104%
robust allowable|0|creep (worst-case limit 0.1)	0.515%	0.485%	0.548%
robust allowable|0.75|creep (worst-case limit 0.03)	0.053%	0.044%	0.064%
robust allowable|0.75|creep (worst-case limit 0.1)	0.571%	0.539%	0.605%
baseline + accumulator 0.75	6.879%	6.769%	6.991%

Table S15. Over-torque probability of every evaluated design on the independent draws with Wilson 95% limits.

Family	Qualifying	In-sample over-torque	In-sample safe	Independent over-torque	95% limits	Independent safe
allowable|0	394	0.66%	49.6%	0.55%	0.52% to 0.58%	49.6%
median|0	293	0.71%	47.4%	0.75%	0.71% to 0.79%	47.6%
allowable|0.75	665	0.61%	58.6%	0.60%	0.57% to 0.64%	58.4%
median|0.75	506	0.77%	57.8%	0.75%	0.72% to 0.79%	57.6%
allowable|0|creep	503	0.81%	57.7%	0.80%	0.76% to 0.84%	57.5%
allowable|0.75|creep	705	0.70%	62.4%	0.70%	0.67% to 0.74%	62.3%
median|0|creep	402	0.80%	60.5%	0.79%	0.75% to 0.83%	60.6%
median|0.75|creep	529	0.73%	61.8%	0.73%	0.69% to 0.77%	61.7%

Table S16. Strict compliance rule: the best design whose in-sample over-torque estimate has an upper Wilson limit of at most 1%, with its in-sample and independent estimates.

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